EXPERIMENT NO: 4

TITLE: Design and Implementation of a Temperature and Pressure Monitoring System Using Industrial Sensors (With DAS mechanism)

AIM: To design and implement a real-time temperature and pressure monitoring system using industrial-grade sensors interfaced with an Arduino microcontroller, integrated with a Data Acquisition System (DAS) for accurate measurement, analysis, and display of process parameters.

APPARATUS:

Sr.No.	Equipment	Specification	Quantity
1	Trainer Kit	Arduino Trainer Kit	1
2	Tinkercad	Online Circuit Simulator	Online Software

THEORY: The design of the system is to measure, monitor, and display temperature and pressure in real time using sensors. By interfacing these sensors with an Arduino microcontroller and a Data Acquisition System (DAS), the system ensures accurate readings, data processing, and visualization for industrial or experimental applications.

Components Overview→

1. **Arduino Microcontroller**
   - Acts as the central processing unit.
   - Reads sensor data, processes signals, and sends data to displays or a computer.
2. **Industrial-Grade Sensors**
   - **Temperature Sensor (e.g., LM35, TMP36, or BMP280)**
     - Measures temperature in Celsius or Kelvin.
     - Provides an analog or digital voltage corresponding to temperature.
   - **Pressure Sensor (e.g., BMP280, MS5611, or MPX5010)**
     - Measures pressure in Pascal (Pa) or bar.
     - Converts physical pressure into electrical signals.
3. **Data Acquisition System (DAS)**
   - Converts analog sensor signals into digital signals if needed.
   - Ensures high-precision data logging.
   - May include additional modules like ADCs for better resolution.
4. **Display Unit**
   - LCD, LED, or OLED screen to display real-time temperature and pressure values.
5. **Tinkercad Simulation**
   - Provides a virtual testing environment to simulate sensors, Arduino, and DAS connections.
   - Allows verification of circuit functionality and coding logic without physical hardware.

Working Principle→

1. **Sensor Reading**
   - Temperature and pressure sensors detect environmental conditions.
   - Sensors generate analog or digital signals proportional to the physical parameters.
2. **Signal Acquisition**
   - Arduino reads analog signals through ADC pins or digital signals through I2C/SPI.
   - DAS ensures accurate measurement, filtering noise, and calibration.
3. **Data Processing**
   - Arduino code converts raw sensor readings into engineering units (°C for temperature, Pa/bar for pressure).
   - Calibration constants are applied for precision.
4. **Data Display**
   - Processed values are displayed on the LCD/OLED.
   - Optional: Data can be transmitted to a computer or cloud for logging.
5. **Simulation in Tinkercad**
   - Connect virtual sensors to Arduino pins in Tinkercad.
   - Write Arduino code using Tinkercad's editor.
   - Observe real-time data readings on the virtual serial monitor or LCD.
   - Helps in debugging and verifying circuit design before hardware implementation.

Advantages→

- Real-time monitoring for industrial and lab applications.
- Accurate and calibrated measurements using DAS.
- Easy debugging and testing using Tinkercad simulation.
- Flexibility to expand system (add more sensors or remote monitoring via Wi-Fi/Bluetooth).

Tinkercad Implementation→

Components Required:

- Arduino UNO
- LCD (16x2)
- TMP36 Temperature Sensor
- FSR (Force Sensitive Resistor / Pressure Sensor)
- 10 kΩ potentiometer (for LCD contrast)
- Jumper wires and breadboard

TMP36 Temperature Sensor:

TMP36 Pin	Connects To	Notes
Left pin (Vs)	5 V	Power
Middle pin (Vout)	A0	Analog input (matches tempPin = A0)
Right pin (GND)	GND	Ground

LCD (16x2) connection:

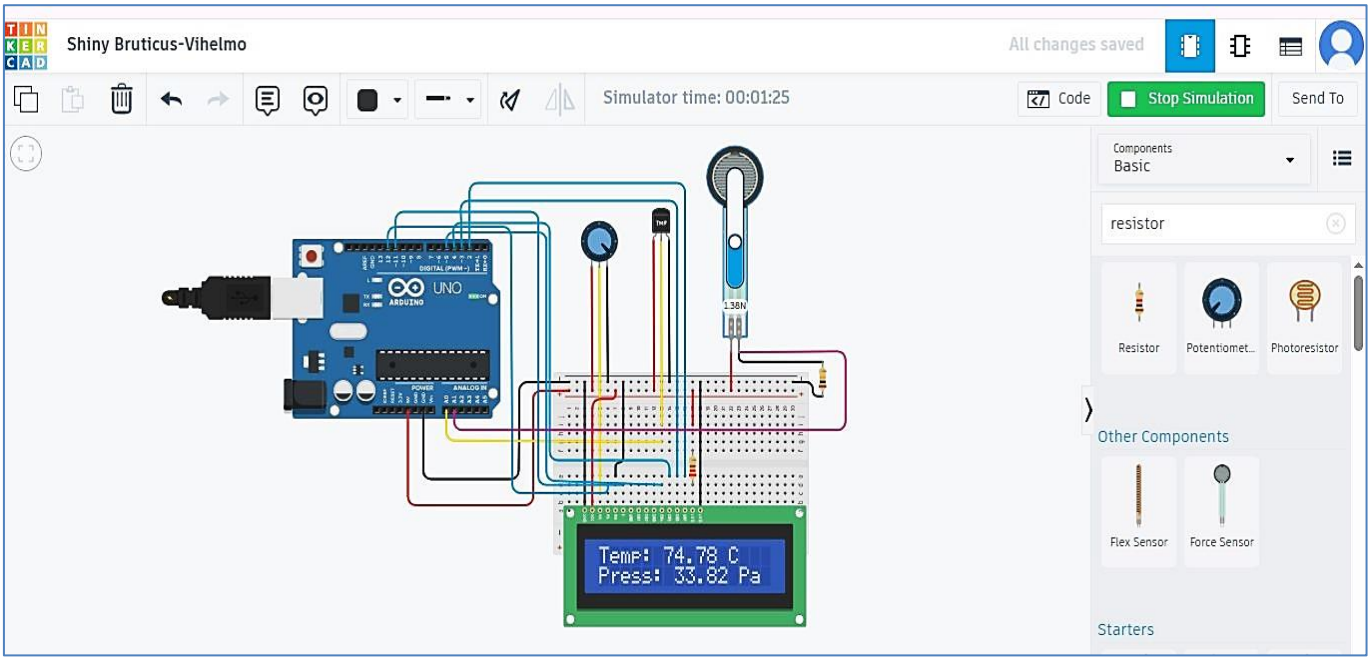
LCD Pin	Function	Arduino Pin	Statement
1 (VSS)	GND	GND	Ground
2 (VDD)	VCC	5 V	Power
3 (V0)	Contrast	Middle pin of potentiometer	Adjust contrast
4 (RS)	Register Select	D12	From code
5 (RW)	Read/Write	GND	Always write mode
6 (E)	Enable	D11	From code
11 (D4)	Data 4	D5	From code
12 (D5)	Data 5	D4	From code
13 (D6)	Data 6	D3	From code
14 (D7)	Data 7	D2	From code
15 (A)	Backlight +	5 V (through 220 Ω resistor)	Optional
16 (K)	Backlight –	GND	Optional

FSR Pressure Sensor:

FSR Pin	Connects To	Notes
One pin	5 V	Power
Other pin	A1 (via 10 kΩ resistor to GND)	Creates a voltage divider for analog reading

Power and Ground:

- Arduino 5V → Breadboard + rail
- Arduino GND → Breadboard – rail
- All sensors and LCD share the same rails.



Simulation Code:

```
// C++ code//
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2); // RS, E, D4-D7
int tempPin = A0; // Temperature sensor
int pressPin = A1; // Pressure sensor
float temperature;
float pressure;
void setup() {
  lcd.begin(16, 2);
  Serial.begin(9600);
  }
void loop() {
  // Read sensors
  int tempVal = analogRead(tempPin);
  int pressVal = analogRead(pressPin);
  // Convert to engineering units (example conversion)
  temperature = (tempVal * 5.0 / 1023.0) * 100; // Celsius
  pressure = (pressVal * 5.0 / 1023.0) * 10; // Arbitrary units
  // Display on LCD
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temperature);
  lcd.print(" C");
  lcd.setCursor(0, 1);
  lcd.print("Press: ");
  lcd.print(pressure);
  lcd.print(" Pa");
  // Print to Serial Monitor
  Serial.print("Temp: ");
  Serial.print(temperature);
  Serial.print(" C, ");
  Serial.print("Press: ");
  Serial.print(pressure);
  Serial.println(" Pa");
  delay(1000);// Update every second
}
```


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```

RESULT:

Put the screenshot of the result of the experiment performed.(Each result page should contain Name, Roll No & Batch Name)

CONCLUSION:

Should be handwritten